

Build sheet — Magnetic Motion drop board. Every measurement needed to build it. The interactive 3D model and drop simulator live in the online version.



BUILD SHEET · PAULGUNA · RAGHAV · JOSE

MAGNETIC MOTION

KEEP MOVING

Board **32 × 44½ in** Field **24 cups** Lean **25°** Price **\$3 / drop**
Middle **15% of drops**

THE BOARD

PYRAMID FIELD, TWO PRIZES

The ball drops through a 3½ in funnel onto a single apex cup, which kicks it hard to one side. From there the pyramid widens row after row. Only four wall wedges — at 13 and 31 in — so the edges stay open and most balls run out to the sides.

That is what makes the middle hard. The four middle bins take up **half the width of the board** and still catch only about 15% of drops. They look easy and they are not.

The **teal centre bin** is the best fidget toy — the one you hang on the board where everyone can see it. It sits right on the 16 in centre line with the guard cup on its edge, so balls get shed off it. About 1 drop in 38 lands there.

Over 25,000 simulated drops: **85% clay ball · 13% fidget toy · 2.6% centre bin**. Zero jams. At 200 plays that is about **169 clay balls and 31 fidget toys**. The simulation is a close guide — cardboard and glue change the bounce, so your 20 real test drops decide it.

PRICE AND PRIZE PLAN

Prize tiers

■ Clay ball – 4 outer bins	85%
■ Fidget toy – 3 blue bins	12%
■ Best fidget toy – teal centre	2.6%

Bring roughly **170 clay balls and 31 fidget toys** for 200 plays. Hang the best fidget toy on the board so people can see what the centre bin pays.

Price: \$3 per drop

High schooler with \$10	3 drops
3rd grader with \$20	6 drops
Realistic plays in 90 min	200–250
Revenue at 220 plays	\$660

Start at \$3. Long line, hold. Standing idle 20 minutes in, swap the card to \$2 and write down the change.

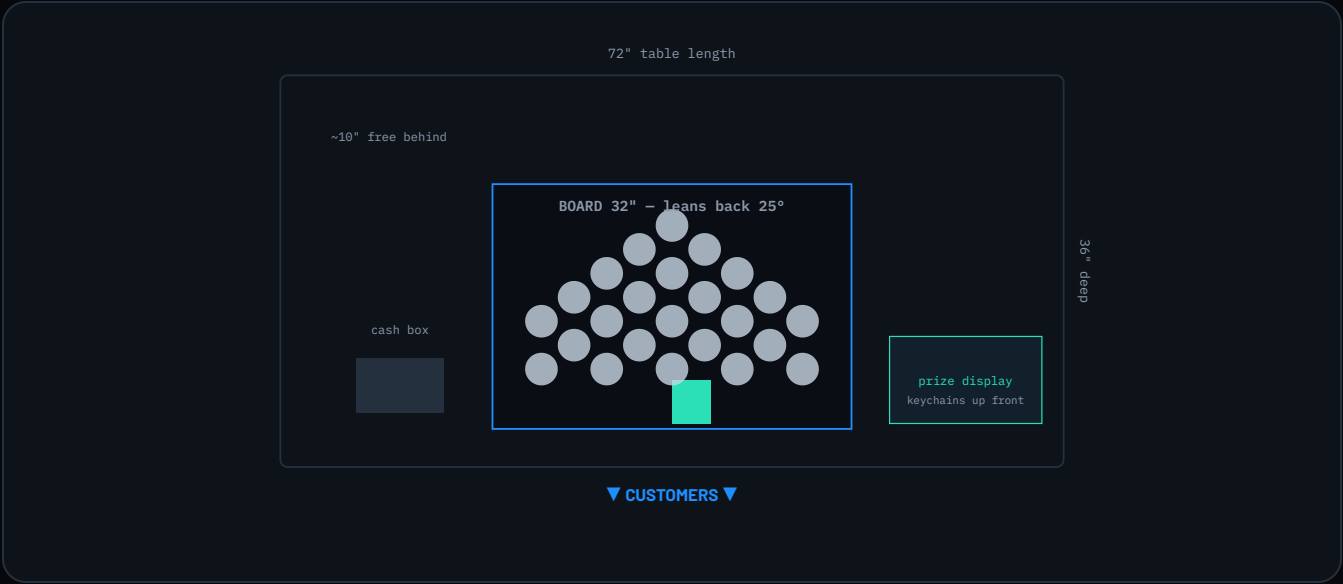
Clay balls have to be dry and hard. A soft or damp clay ball marks hands, tables and gym floors, and it is the prize 85% of your customers walk away with. If they are not fully cured, swap the base prize for anything dry you have in bulk – the odds do not change.

Third graders can't reach the funnel. The top sits about 68 in off the floor. Bring a milk crate or step stool – those are your \$20 customers.

YOUR TABLE

LAYOUT ON THE FOLDING TABLE

Board on the front edge, centered, leaning back 25°. Drawn for a 72 × 36 in table.



MATERIALS

WHAT YOU NEED TO BUILD IT

Almost nothing left to buy — you already own the cups, the cardboard and all four prize tiers. Budget for the build is about a dollar.

Still to buy

Ping-pong balls, 6-pack — orange shows up better than white on a black board	\$1.25
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Hot glue sticks, only if you're out	\$0–1.25
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Black paint or a wide black marker for the backboard, if you have none	\$0–1.25
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Build total	about \$1.25–3.75
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Already yours

Backboard 32 × 44½ — already cut	✓
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Side walls 4 × 44½ — already cut	✓ ×2
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Party cups — you have 50, the board uses 24	24 of 50
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Clay balls — the base prize	~170
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Small fidget toys	~31
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Cardboard

Your backboard and both side walls are already cut. What is left — funnel walls, wedges, bin floor, lip, 7 dividers, 2 braces, crossbar, sign and supports — comes to about **11 sq ft**. Two or three flattened moving boxes cover it.

The braces are the biggest remaining pieces: two right triangles **12⁵/₈ × 27¼ in.** Cut those first, while your cardboard is still large.

Tools and tape

Hot glue gun

Box cutter or strong scissors — used at home only

Yardstick, pencil, marker

Duct tape (brace hinges) and packing tape

Painter's tape to hold the board to the table

Binder clips for the crossbar

Velcro dots for the swappable price card

Milk crate or step stool for younger players

On the \$10 cap. The build itself lands near a dollar, but the prizes you already ordered — 250 bouncy balls, 36 cubes, 20 keychains, 8 magnet sets — are well past \$10 of real money. Report the true figure in your Reflection Assignment rather than the cap; the grade is on understanding your real profit, and \$660 of revenue against a \$10 cost won't survive a question.

CUT LIST

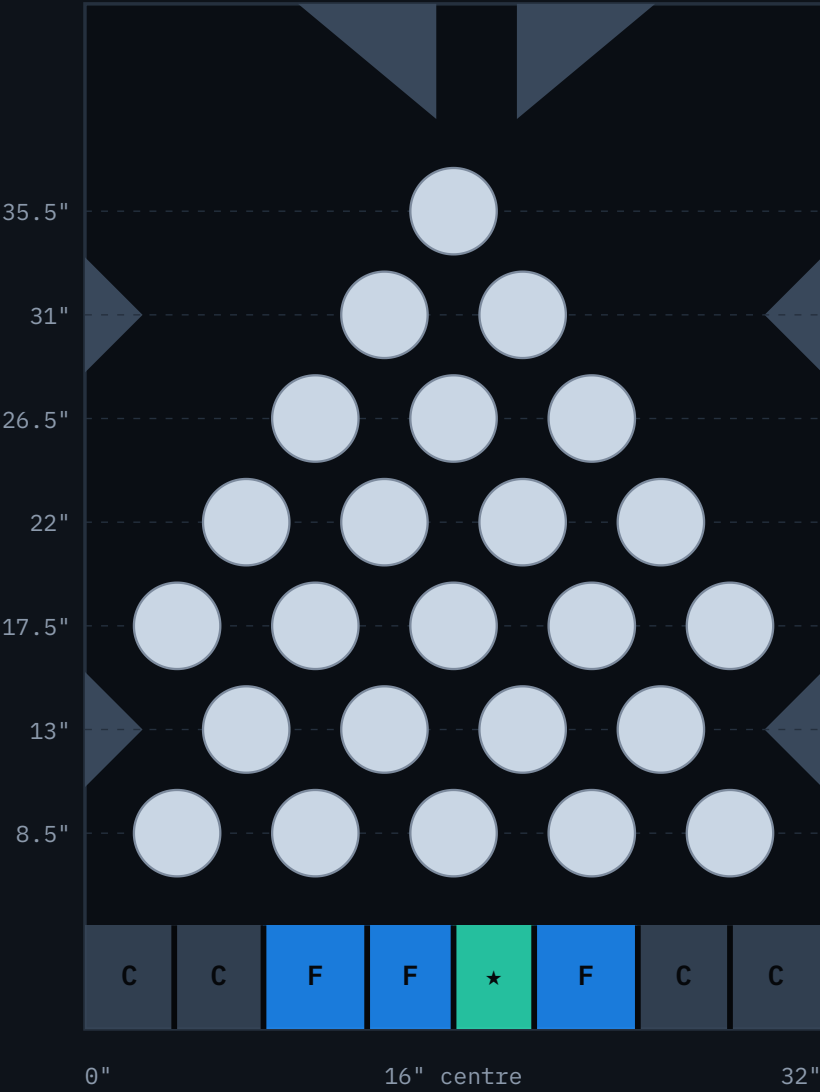
EVERY CARDBOARD PIECE

All inches. Run the corrugation up and down on the backboard and walls — much stiffer that way. Cut everything at home; box cutters aren't allowed on campus.

PIECE	SIZE (IN)	QTY	JOB
A Backboard	32 × 44½	1	Yours already. Paint it black before anything else goes on.
B Side walls	4 × 44½	2	Yours already. 4 in is plenty — the ball sits only ¾ in off the board.
C Funnel walls	4 × 8	2	Angled at the top, tips at 14¼ and 17¾ in — a 3½ in gap centred on 16. Sends every ball onto the apex cup.
K Wall wedges	4 × 7⅛	4	Fold each in half into a V and glue to the walls, at 13 in and 31 in only. Four total — adding more would feed the middle and make it too easy.
D Bin floor	32 × 4	1	Shelf the ball lands on.
E Front lip	32 × 3	1	Front of the bins. Print or letter the tiers on it.
F Bin dividers	4 × 4½	7	Splits the bottom into 8 bins. The 7th is the one that separates the keychain bin from the magnet bin.
G Back braces	right triangle 12⅝ × 27¼	2	Holds the 25° lean on the taller board.
H Crossbar	20 × 2	1	Ties the braces together.
I Header sign	32 × 8	1	Name, slogan, price. Print file is 4800 × 1200 px.
J Sign supports	2 × 12	2	Hold the sign above the board.

WHERE EVERY CUP GOES

Measure from the **bottom-left corner** of the backboard. Each number is the **center** of a cup. Rows are 4.6 in apart going up.



ROW	UP FROM BOTTOM	CUP CENTERS FROM LEFT
1 (top)	35.5"	16" apex cup – every ball hits this first
2	31"	13" · 19"
3	26.5"	10" · 16" · 22"
4	22"	7" · 13" · 19" · 25"

ROW	UP FROM BOTTOM	CUP CENTERS FROM LEFT
5	17.5"	4" · 10" · 16" · 22" · 28"
6	13"	7" · 13" · 19" · 25"
7 (bottom)	8.5"	4" · 10" · 16" · 22" · 28" 16" is the guard cup

Bin dividers, from left edge

$3\frac{7}{8}" \cdot 7\frac{3}{4}" \cdot 12\frac{3}{4}" \cdot 16" \cdot 19\frac{1}{2}" \cdot 24" \cdot 28"$

Dividers are 5 in tall and 5 in deep. These numbers change with the two sliders above.

Before you glue

Measure across a cup rim. Spacing is 6 in, so rims must be **3.8 in** or less — standard party cups are about $3\frac{3}{4}$ in, which just fits. You need **24 cups**, so your 50 leaves plenty of spares.

STEP BY STEP

About 3 hours with two people.

- 1 Cut every piece** from the cut list, at home. Label each with its letter in pencil.
- 2 Paint the backboard black** and let it dry fully before anything else.
Black is the whole look — silver cups and a blue center read hard against it, and it matches the logo. Paint is miserable to do once the cups are on.
- 3 Mark the peg map.** Pencil lines across at $8\frac{1}{2}$, 13, $17\frac{1}{2}$, 22, $26\frac{1}{2}$, 31 and $35\frac{1}{2}$ in up from the bottom — every $4\frac{1}{2}$ in, seven rows. Then mark each cup centre along its row.
Set a cup rim-down on each mark and trace it. Now you have a circle for every cup.
- 4 Glue the side walls (B)** along both edges, standing straight out 4 in.
- 5 Glue the cups upside-down** — rim on the board, closed bottom facing out. Full ring of hot glue on the rim, press onto the traced circle, hold 20 seconds.
Upside-down matters. If the open end faces out, a ball can drop inside and stick.
- 6 Glue the funnel walls (C)** at the top, angled inward, tips at **$14\frac{1}{4}$ and $17\frac{3}{4}$ in**, bottom edge at $39\frac{1}{2}$ in. That leaves a $3\frac{1}{2}$ in gap centred on 16.
Keep the gap at $3\frac{1}{2}$ in and the apex cup at $35\frac{1}{2}$ in. Any closer together and the ball wedges between them and stops dead.
- 7 Glue the wall wedges (K)** at 13 in and 31 in up, both walls, sticking **$2\frac{1}{2}$ in** inward. Four total — do not add a pair at 22 in, that is what keeps the middle hard.
These matter more than they look. Without them the ball finds a clear channel down each side and 84% of drops land in the two end bins.
- 8 Build the bins.** Bin floor (D), then the 7 dividers (F) at the positions in the peg map, then the front lip (E).
Tier labels, left to right: **CLAY · CLAY · FIDGET · FIDGET · ★ BEST · FIDGET · CLAY · CLAY**. Dividers go at **$3\frac{7}{8}$ · $7\frac{3}{4}$ · $12\frac{1}{4}$ · 16 · $19\frac{1}{2}$ · 24 · 28 in** from the left edge.
- 9 Attach the braces (G)** at 6 in and 26 in from the left, with duct-tape hinges rather than glue so they fold flat for the car. Clip the crossbar (H) across the bottom.
Check the lean: standing up, the top of the board should sit about **$18\frac{3}{4}$ in** behind the bottom edge.
- 10 Mount the header sign (I, J)** so the name clears the top of the board.
- 11 Drop-test 20 balls.** You want about 17 landing in the clay bins and 3 in the middle four. The teal centre bin is rare by design — roughly 1 drop in 38.

IF THE TEST DROPS GO WRONG

WHAT YOU SEE	FIX
Too many fidget toys going out	Narrow all four middle bins by $\frac{1}{2}$ in each. The outer clay bins widen automatically and soak up the difference.
Almost nobody reaches the middle	Widen the centre bin toward $4\frac{1}{2}$ in, or take one pair of wall wedges off.
Zero magnets or keychains in 40 drops	Pull off the guard cup at 17 in on the bottom row. That alone roughly doubles both rates.
Nearly everything lands in the two end bins	A wall wedge is missing or too small. All six must stick $2\frac{1}{2}$ in inward.
Ball flies off the front	Lean it back to about 30° — top sits 21 in behind the bottom edge.
Ball sticks between cups	A glue blob is narrowing the gap. Scrape it. Every gap has to fit two fingers side by side.
Ball jams under the funnel	The funnel is too close to the apex cup. Gap must be $3\frac{1}{2}$ in and the apex cup no higher than $35\frac{1}{2}$ in.
Board tips or slides	Painter's tape the braces and bin floor to the table, textbook on the crossbar.

RUNNING IT WITH NO CHAIRS

Positions

Front, beside the board: takes \$3, hands the ball, calls the bin, hands the prize

Behind the table: cash box and restocking

Out front pitching — rotate every 20 minutes

Bring

Milk crate or step stool for the younger kids

All your balls — they roll away and get stepped on

Glue gun and 5 spare cups for repairs

Painter's tape, binder clips, duct tape

Tally sheet: every drop, which bin, which prize